College of Engineering – Department of Chemical Engineering

CURRICULUM

Bachelor of Science in Chemical Engineering

Academic Year 2025-2026

Reference CMOs: CMO N0. 91 s. 2017, CMO No. 20, s. 2013, CMO No. 4 s. 2018 and CMO No. 39 s. 2021, CMO No. 40 s. 2021

Curriculum Description

Chemical Engineering is a profession that involves the conceptualization, development, design, improvement and application of safe, healthy, ethical and economical way of utilizing materials and energy in unit Processes and operations for the benefit of society and the environment through the knowledge of mathematics, chemistry, biology, information technology and other natural, applied and social sciences, gained by study, research and practice

Program Education Objectives

The chemical engineering alumni three to five years after graduation shall:
PEO 1: Successfully practice as defined in the law governing the scope of Chemical Engineering.
PEO 2: Assume leadership position in industry, academe, government or private sector with consideration to social and ethical responsibility.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.
SO7 Communication. Communicate effectively on complex engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;
SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

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SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.
SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.
SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex engineering activities.
SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.
SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
MATH 101 - Differential Calculus	4	0	4
MATH 102 - Integral Calculus	4	0	4
AI 102 - Probability and Statistics for AI	3	0	3
MATH 103 - Differential Equations	3	0	3
Subtotal	**14**	**0**	**14**
B. Natural/ Physical Sciences			
CHEM 101 - Chemistry for Engineers	3	3	4
BioT 101 - Biotechnology	3	0	3
PHYS 101 - Physics for Engineers	3	3	4
Subtotal	**9**	**6**	**11**
C. Basic Engineering Sciences			
ENGG 101 - Introduction to Engineering	0	3	1
ENGG 102 - Engineering Drawing	0	3	1
CpE 101 - Computer Programming	0	3	1
ENGG 103 - Computer-Aided Design	0	3	1
ENGG 104 - Engineering Economics	3	0	3
ENGG 106 - Engineering Management	2	0	2
Subtotal	**5**	**12**	**9**
D. Allied Courses			
CHEM 201 - Analytical Chemistry	4	3	5
CHEM 202 - Organic Chemistry	4	3	5
EE 213 - Basic Electrical and Electronics Engineering	2	3	3
ENGG 105 - Technopreneurship	3	0	3
ENGG 109 - Engineering Mechanics	3	0	3
ENGG 113 - Environmental Science and Engineering	3	0	3
ENGG 118 - Material Science and Engineering	3	0	3
IE 405 - Qualitative Methods in Management	3	0	3
Subtotal	**25**	**9**	**28**
E. Professional Courses			
CHE 201 - Chemical Engineering Calculations	2	3	3

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CHE 202 - Physical Chemistry for ChE 1	2	3	3
CHE 203 - Advanced Engineering Mathematics for ChE	3	0	3
CHE 204 - Chemical Engineering Thermodynamics	2	3	3
CHE 205 - Momentum Transfer	2	3	3
CHE 206 - Chemical Process Laboratory	0	3	1
CHE 207 - Chemical Process Industries	3	0	3
CHE 301 - Heat and Mass Transfer	3	3	4
CHE 302 - Computer Applications in Chemical Engineering	0	3	1
CHE 303 - Process Dynamics and Control	2	3	3
CHE 304 - Particle Technology	2	3	3
CHE 305 - Physical Chemistry for ChE 2	2	3	3
CHE 306 - Solution Thermodynamics	3	0	3
CHE 307 - Chemical Engineering Laboratory 1	0	3	1
CHE 308 - Separation Processes	2	3	3
CHE 309 - Chemical Reaction and Engineering	3	3	4
CHE 310 - Industrial Waste Management and Control	3	0	3
CHE 311 - Chemical Engineering Project 1	1	3	2
CHE 312 - Track Specialization 1	3	0	3
CHE 313 - Chemical Engineering Laboratory 2	0	3	1
CHE 314 - Plant Visits and Seminar	0	3	1
CHE 315 - Track Specialization 2	3	0	3
CHE 401 - Chemical Engineering Design 1	1	3	2
CHE 402 - Chemical Engineering Project 2	0	3	1
CHE 403 - Biochemical Engineering	3	0	3
CHE 404 - Laws, Ethics and Process Safety for ChE	2	0	2
CHE 405 - Track Specialization 3	3	0	3
CHE 406 - Chemical Engineering Application with Comprehensive Examination	0	6	2
CHE 407 - Chemical Engineering Design 2	2	3	3
ENGG 116 - Methods of Research	3	0	3
ENGG 117 - On-the-Job Training		320	4
Subtotal	**55**	**383**	**80**
TOTAL TECHNICAL COURSES	**108**	**410**	**142**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
GEd 101 - Understanding the Self	3	0	3
GEd 102 - Mathematics in the Modern World	3	0	3
GEd 104 - The Contemporary World	3	0	3
GEd 105 - Readings in Philippine History	3	0	3
GEd 106 - Purposive Communication	3	0	3
GEd 107 - Ethics	3	0	3
GEd 108 - Art Appreciation	3	0	3

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GEd 109 - Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives			
AI 101 - Linear Algebra for AI	3	0	3
AI 103 - Machine Learning and Neural Networks	2	3	3
GEdE 101 - Entrepreneurial Mind	3	0	3
GEd 103 - Life and Works of Rizal	3	0	3
Subtotal	**11**	**3**	**12**
C. Mandated Courses			
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
PATHFit 4 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
NSTP 111 - National Service Training Program 1	3	0	3
NSTP 121 - National Service Training Program 2	3	0	3
Subtotal	**14**	**0**	**14**
TOTAL NON-TECHNICAL COURSES	**49**	**3**	**50**
GRAND TOTAL	**157**	**413**	**192**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**14**
B. Natural/Physical Sciences	**11**
C. Basic Engineering Sciences	**9**
D. Allied Courses	**28**
E. Professional Courses	**80**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives	**12**
C. Mandate Courses	**14**
GRAND TOTAL	**192**

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CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN CHEMICAL ENGINEERING
CMO N0. 91 s. 2017, CMO No. 4 s. 2018, CMO No. 20 s. 2013, CMO No. 39, s. 2021 and CMO 40, s. 2021

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHEM 101	Chemistry for Engineers	4	3	3			TC-B
ENGG 101	Introduction to Engineering	1	0	3			TC-C
ENGG 102	Engineering Drawing	1	0	3			TC-C
GEd 102	Mathematics in the Modern World	3	3	0			NTC-A
GEd 104	The Contemporary World	3	3	0			NTC-A
GEd 105	Readings in Philippine History	3	3	0			NTC-A
GEd 108	Art Appreciation	3	3	0			NTC-A
MATH 101	Differential Calculus	4	4	0			TC-A
NSTP 111	National Service Training Program 1	3	3	0			NTC-C
PATHFit 1	Movement Competency Training	2	2	0			NTC-C
	Total	**27**	**24**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 101	Computer Programming	1	0	3			TC-C
GEd 101	Understanding the Self	3	3	0			NTC-A
GEd 106	Purposive Communication	3	3	0			NTC-A
GEd 109	Science, Technology and Society	3	3	0			NTC-A
GedE 101	Entrepreneurial mind	3	3	0			NTC-B
MATH 102	Integral Calculus	4	4	0	MATH 101		TC-A
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC-C
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC-C
PHYS 101	Physics for Engineers	4	3	3	MATH 101		TC-B
	Total	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
BioT 101	Biotechnology	3	3	0	CHEM 101		TC-B
GEd 103	Life and Works of Rizal	3	3	0			NTC-C
GEd 107	Ethics	3	3	0			NTC-A
	Total	**9**	**9**	**0**			

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SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 101	Linear Algebra for AI	3	3	0			NTC-B
CHE 201	Chemical Engineering Calculations	3	2	3	CHEM 101		TC-E
CHEM 201	Analytical Chemistry	5	4	3	CHEM 101		TC-D
CHEM 202	Organic Chemistry	5	4	3	CHEM 101		TC-D
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		TC-C
ENGG 109	Engineering Mechanics	3	3	0	PHYS 101		TC-D
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities 1	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**25**	**21**	**12**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC-A
CHE 202	Physical Chemistry for ChE 1	3	2	3	MATH 103, CHEM 201		TC-E
CHE 203	Advanced Engineering Mathematics for ChE	3	3	0	MATH 103		TC-E
CHE 204	Chemical Engineering Thermodynamics	3	2	3	CHE 201		TC-E
CHE 205	Momentum Transfer	3	2	3	MATH 103, ChE 201		TC-E
CHE 206	Chemical Process Laboratory	1	0	3	CHEM 202		TC-E
CHE 207	Chemical Process Industries	3	3	0	CHEM 202		TC-E
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		TC-D
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities 2	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**24**	**20**	**12**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHE 301	Heat and Mass Transfer	4	3	3	CHE 205		TC-E
CHE 302	Computer Applications in Chemical Engineering	1	0	3	CpE 101, CHE 203		TC-E
CHE 303	Process Dynamics and Control	3	2	3	CHE 203, ENGG 103		TC-E
CHE 304	Particle Technology	3	2	3	CHE 205		TC-E

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CHE 305	Physical Chemistry for ChE 2	3	2	3	CHE 202		TC-E
EE 213	Basic Electrical and Electronics Engineering	3	2	3	PHYS 101		TC-D
ENGG 104	Engineering Economics	3	3	0	Third Year Standing, AI 102		TC-C
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101		TC-D
ENGG 116	Methods of Research	3	3	0	3rd Year Standing, AI 102		TC-E
	Total	**26**	**20**	**18**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CpE 101		NTC-B
CHE 306	Solution Thermodynamics	3	3	0	CHE 203 CHE 204, CHE 302		TC-E
CHE 307	Chemical Engineering Laboratory 1	1	0	3	CHE 205, CHE 301	CHE 308	TC-E
CHE 308	Separation Processes	3	2	3	CHE 204, CHE 301		TC-E
CHE 309	Chemical Reaction and Engineering	4	3	3	CHE 301, CHE 305, CHE 203		TC-E
CHE 310	Industrial Waste Management and Control	3	3	0	ENGG 113, CHE 304		TC-E
CHE 311	Chemical Engineering Project 1	2	1	3	ENGG 116		TC-E
CHE 312	Track Specialization 1	3	3	0	Third Year Standing		TC-E
ENGG 106	Engineering Management	2	2	0	ENGG 104		TC-C
	Total	**24**	**19**	**15**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHE 313	Chemical Engineering Laboratory 2	1	0	3	CHE 307		TC-E
CHE 314	Plant visits and seminars	1	0	3	3rd Year Standing, CHE 207		TC-E
CHE 315	Track Specialization 2	3	3	0	CHE 312		TC-E
	Total	**5**	**3**	**6**			

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FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHE 401	Chemical Engineering Design 1	2	1	3	Fourth year Standing, CHE 308, CHE 304, CHE 309		TC-E
CHE 402	Chemical Engineering Project 2	1	0	3	CHE 311		TC-E
ENGG 105	Technopreneurship	3	3	0	Fourth Year Standing		TC-D
ENGG 117	On-the-Job Training	4		320	Fourth year standing		TC-E
	Total	**10**	**4**	**326**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHE 403	Biochemical Engineering	3	3	0	CHE 309, CHEM 202		TC-E
CHE 404	Laws, Ethics and Process Safety for ChE	2	2	0	Fourth year standing		TC-E
CHE 405	Track Specialization 3	3	3	0	CHE 315		TC-E
CHE 406	Chemical Engineering Application with Comprehensive Examination	2	0	6	Graduating		TC-E
CHE 407	Chemical Engineering Design 2	3	2	3	CHE 401, ENGG 106, CHE 303		TC-E
IE 405	Quantitative Methods in Management	3	3	0	ENGG 104, ENGG 105		TC-D
	Total	**16**	**13**	**9**			